

Hurdles Log — Kushige 2013 vs. Ito 2009 rhythmic-gene comparison

This file is the deliverable. Mike wants the analysis hurdles characterized, because whatever breaks on this data will break on the lab's own across-organism metagenomic data. Keep this open in a tab while you work and append to the ENCOUNTERED section the moment something bites — don't reconstruct it at the end. The friction IS the data; the grubby dataset-specific snags you'd otherwise smooth over are the most valuable entries.

For each encountered hurdle, three things: **what bit**, **what you did**, and **does it transfer** to env-vs-reference and/or env-vs-env (MAG-to-MAG). That third line is what turns a lab notebook into what Mike asked for.

Scope reminder: this run is **Version A** (compare orthologs among the RHYTHMIC genes; amplitude in ranks; phase as a dawn/dusk 2x2). Version B (all-orthologs, unbiased) is phase two — the full datasets ARE available, so it's a real option, but not on the few-days timeline.

PART 1 — PREDICTED HURDLES (from the project brief)

These were anticipated before opening the files. Update each **Status:** line as you actually hit it — confirm, revise, or note it didn't materialize. Predictions being wrong is itself worth recording.

H1 — Rhythmicity-threshold selection bias [DEMONSTRATED]

Conditioning a correlation on "rhythmic in dataset X" is selection on a noisy variable -> range restriction + regression to the mean -> correlation biased down, plus a spurious "Anabaena is less rhythmic" story. Version A IS the biased version by construction; demonstrate the trap by running it and noting that an all-orthologs analysis (Version B) removes it. Transfer: recurs identically the first time the lab compares rhythmic genes across its own conditions. HIGHEST transfer value. **Status:** *confirmed, sharply* — see 2026-07-27 entries below. Restricting to either paper's own headline rhythmic set collapses the name-matched ortholog overlap from an ad hoc 16 genes down to 1-2, stable across the entire 50th-90th amplitude-percentile range. Selection on rhythmicity doesn't just bias a correlation here, it nearly empties the comparison set outright. Version B (all-RBH-orthologs) mostly fixes this -- 565 usable pairs, no two-sided rhythmicity gate -- but see the "Version B built" entry for a residual ONE-SIDED version of H1 that survived anyway, driven by data publication, not methodology: Ito2009 only reports a peak time for genes it calls cycling at all.

H2 — Orthology mapping [DEMONSTRATED, easy instance]

5,368 vs. ~2,700 genes, paralog families in Anabaena. Default RBH (clean, 1:1, discards paralogs). FALLBACK to protect timeline: name-match on gene symbols both papers print (kaiABC, apcE, pecB, groEL...) — for Version A you only need orthologs among the rhythmic sets, so name-matching is a legitimate primary method here. Split transfer: env-vs-reference = well rehearsed (ambiguity one-sided). env-vs-env (MAG-to-MAG) = only partially rehearsed and strictly harder — no ground truth either side, and MAG incompleteness is non-random (GC/coverage skewed) so "gene absent" correlates with gene

class correlates with rhythmicity class = a confound shaped like biology, invisible to RBH. Do NOT read a clean result here as evidence the method survives two MAGs. **Status:** *name-match used FIRST (no proteome FASTAs in the repo yet; RBH looked like its own sub-project)*. See 2026-07-27 "Version A" entry — *name-matching's recall turned out to be a real limiter, not just a documented caveat: only ~25-50% of hits on either side carry a gene symbol at all. Went back and actually built RBH afterward (see "Version B built" entry) -- turned out NOT to be the multi-day swamp the brief warned about: both reference genomes still carry their classic locus tags as GFF **old_locus_tag** (96.5% / 90.4% coverage), so the ID-reconciliation step that usually makes RBH painful was clean. 1,846 genome-wide RBH pairs vs. name-matching's 1-2. Lesson: the brief's own fallback advice ("use RBH only if it's already familiar and quick") undersold how quick it actually was once tried -- worth trying RBH first, not last, when both genomes are this well-curated.*

H3 — CV as a mean-dependent, cross-platform amplitude measure [DEMONSTRATED]

"Amplitude" = CV = SD/mean, inflates for low-expression genes; two different arrays so absolute CVs not comparable. Mitigation: compare in RANKS (Spearman), not raw CV. Transfer: recurs, likely worse, on coverage-dependent metagenomic data. **Status:** *confirmed and quantified — see 2026-07-27 entry. The shared numeric anchor $10^{-0.6}$ (~0.251) that both papers use in their threshold definitions sits at the 90th percentile of Ito2009's own amplitude distribution but only the 19th percentile of Kushige2013's (Kushige's raw CVs run ~2.7x higher across the board: median 0.341 vs 0.124). The "same number" is not remotely the same stringency on the two platforms.*

H4 — Attenuation of correlation by measurement noise [DESCRIBED — not executed]

Both amplitudes noisy -> any r attenuated. In principle: estimate reliability from replicate–replicate correlation (both $n=2$) and disattenuate. NOT done on this timeline (needs per-replicate values that may not be in the SIs; it's a mini-project). Named as the correction a rigorous Version B would need. Transfer: recurs whenever replication is thin — i.e. essentially always in env sampling. **Status:** *described only — but now that you have the full files, note whether per-replicate columns are actually present (that determines if it's doable in phase two)*

H5 — Phase is circular [DEMONSTRATED]

Peak time wraps at ~24 h. Do NOT Pearson peak times. Version A uses a dawn/dusk 2x2 contingency table (Fisher + odds ratio). Fuller circular stats (Rayleigh, mean resultant length, circular correlation coefficient) are in Schmelling 2017 if wanted. Transfer: recurs identically. **Status:** *pending*

H6 — FDR / rhythmicity-call calibration [DESCRIBED — not executed]

Both papers' rhythmicity = cosine-correlation P on a short, autocorrelated series -> effective df below timepoint count -> P anticonservative; amplitude thresholds stringent; gene counts (78, 97) are loose-test x tight-filter products. Use the papers' published calls as-is. Re-calling with JTK_CYCLE / RAIN is parked. Transfer: named as the caller-choice decision the lab faces for its own data. **Status:** *using published calls as-is*

H7 — Data provenance / recoverability [DEMONSTRATED — it already bit]

Manifested during scoping: SIs unverifiable from an automated context; Kushige's raw arrays in a dead database (KEGG Expression DB, ex0001892-ex001947). NOTE: both full SIs were successfully

downloaded manually — so the gate cleared, but the lesson stands. Lesson for the lab's pipeline: archive every intermediate per-gene table yourself, durable and machine-readable — journal SIs and project databases are not reliable five-year stores. Transfer: recurs as "did our pipeline preserve the intermediates we'll need later." **Status:** *confirmed — full tables obtained by hand; log the source URLs + dates in data/raw/SOURCES.txt*

PART 2 — ENCOUNTERED HURDLES (append live)

The valuable part. Every time something is annoying, ambiguous, or forces a judgment call, add an entry. Template below — copy it, don't overthink it, one line each.

2026-07-27 — Properly-scoped Version A overlap is too small to support phase statistics, at any threshold

What bit: An earlier ad hoc pass (matching gene names across the FULL 2515/5336-gene tables using each paper's loosest own "cycling vs. AR" call) produced a workable-looking 16-gene overlap and a flashy result (narM near-antiphase between species). That set does NOT correspond to either paper's actual headline rhythmic definition:

- Kushige2013's "78 cycling genes" (Table S2, in the SI PDF -- NOT in the Data Set S1 spreadsheet, which has no amplitude/P columns at all, only raw time series) = tier a+b under N+ ($P < 0.01-0.05$, $\text{amplitude} > 10^{-0.6}$ to $10^{-0.7}$). The looser tier c (+114 more genes) is explicitly excluded from the paper's own "78" figure.
- Ito2009's "97 higher-amplitude genes" (per the SI PDF's Table S2 footnote, not tabulated by name anywhere in the SI) = $P < 0.001$ AND $\text{amplitude} > 10^{-0.6}$ in the wild-type. Filtering Dataset S1 directly on those numeric columns reproduces 95, not exactly 97 -- close enough to confirm the threshold, off by a small rounding/boundary-inclusivity gap not worth chasing further.

Re-matching gene names between these two PROPERLY-scoped sets: only **2 pairs** survive ([gap1](#), [kaiB](#)). narM is in neither set -- it's Kushige tier c (excluded from the 78) and its Ito values ($P = 0.017$, $\text{amplitude} = 0.117$) are nowhere near the 97-gene bar. The 16-gene "finding" doesn't survive contact with the papers' own definitions.

Suspecting the amplitude thresholds themselves weren't comparable across platforms (H3), checked directly: the shared numeric anchor $10^{-0.6}$ sits at the 90th percentile of Ito's full amplitude distribution but only the 19th percentile of Kushige's (Kushige's raw CVs run $\sim 2.7\times$ higher across the board -- different array platform, different noise floor). Re-derived both rhythmic sets by AMPLITUDE PERCENTILE RANK within each platform's own distribution instead of a shared raw number (P-value criteria left as each paper's own). Result: WORSE, not better -- 0 overlapping names at the (matched) 90th percentile. Swept the whole percentile range as a sensitivity check instead of cherry-picking one number:

percentile	Ito n (named)	Kushige n (named)	name-matched overlap
90th	96 (44)	105 (28)	0
85th	102 (45)	159 (40)	1 (gap1)
80th	107 (47)	197 (52)	1 (gap1)

percentile	Ito n (named)	Kushige n (named)	name-matched overlap
75th	111 (50)	243 (63)	1 (gap1)
70th	114 (52)	305 (77)	1 (gap1)
60th	117 (53)	412 (99)	2 (gap1, rpaC)
50th	122 (56)	508 (114)	2 (gap1, rpaC)

Overlap sits at n=1-2 across the ENTIRE 50th-90th range. Not a threshold-calibration problem -- moving the cutoff doesn't rescue it. n=1-2 cannot support a circular phase-offset histogram, a Rayleigh non-uniformity test, or an amplitude-vs- $|\Delta\phi|$ correlation; none of those were built.

What I did: Built the actual `data/` pipeline the brief specifies (`src/01_parse_kushige.py`, `02_parse_ito.py` -- parsing the papers' own curated tables from the SI PDFs via `pdftotext -layout`, not recomputing from raw arrays where avoidable; `03_orthologs_namematch.py`, `04_percentile_rhythmic_sets.py`), with every ortholog row explicitly flagged `ortholog_method="name-match"`, `rbh_confirmed=False` rather than letting that distinction get silently lost. Stopped short of forcing a percentile choice that "works" -- reported the sensitivity table instead and let the small-n result stand as the finding. One process note: `pdftotext`-extracted text from a PowerPoint-derived PDF contained stray `\r` characters from overstruck text that Python's universal-newline `readlines()` silently split on (but `wc/grep` don't), desynchronizing line-indexed parsing by ~30 lines with no error -- fixed by reading bytes and splitting on `\n` only. Also: writing scratch files to plain `/tmp` in this environment was unstable (content differed between back-to-back reads of the same path) - moved to the session scratchpad dir instead.

Open question, not resolved: whether the small overlap reflects real biology (Kushige's highest-amplitude genes are heavily nitrogen-fixation/heterocyst-related -- `nifB/nifS/nifX/nifV1/ctaC/hupC` -- and phycobilisome genes; Ito's are core clock machinery, transporters, and phototaxis genes -- almost disjoint functional programs) or is a name-matching recall artifact (only ~25-50% of hits on either side carry a gene symbol at any percentile; a real ortholog with an unnamed partner on either side is invisible to this method). Cannot be resolved without RBH.

Transfers?: Both directions, and this is probably the single highest-value entry in this register for that reason. env-vs-reference: identical mechanism -- if the lab's own rhythmicity threshold and the reference paper's threshold aren't calibrated to the same percentile of each platform's/method's own noise distribution, any "shared rhythmic set" comparison is comparing differently-selected populations before any biology enters, and a small resulting n may look like a clean negative result when it's actually an artifact of the intersection procedure. env-vs-env (MAG-to-MAG): recurs and compounds -- two non-reference genomes means the H3 percentile-mismatch problem AND the H2 name-matching recall problem both apply on both sides simultaneously, with no anchor to calibrate against. Recommend for the lab's pipeline: (1) never compare two platforms' rhythmicity calls on shared raw thresholds without checking where those thresholds land in each platform's own full distribution first; (2) treat a small name-matched overlap as ambiguous (biology vs. method-recall) by default, not as a negative result, unless RBH or equivalent backs it up.

What bit: After the Version A writeup above, decided to actually build Version B rather than just recommend it. Fetched both organisms' own reference-genome proteomes from NCBI (RefSeq GCF_000012525.1 for 7942, GCF_000009705.1 for Anabaena/Nostoc PCC 7120 -- the exact genomes each paper's own locus tags come from). Confirmed the ID chain is clean before committing: GFF `old_locus_tag` -> shared gene `ID` -> CDS `protein_id` -> FASTA header, on both genomes, no ambiguous (duplicated) `old_locus_tag` mappings. Ran reciprocal `blastp` (evalue 1e-5, best hit by bitscore, not `-max_target_seqs` which isn't a strict best-hit guarantee) -> **1,846 RBH ortholog pairs**, no rhythmicity filter anywhere in that step.

Joining those pairs against full per-gene tables (not just the rhythmic subsets) hit an asymmetry that only showed up at execution time: Ito2009's spreadsheet has a literal "-" placeholder instead of a peak time for every gene it doesn't call cycling (1,715 of 2,515 genes) -- there's no raw time series in Ito's public data to fall back on and compute an alternative phase estimate the way Kushige's data allows (Kushige2013 DOES publish per-timepoint raw values for all 5,336 genes, letting a peak-time proxy be computed for every one of them regardless of significance). Dropping rows with an undefined peak time therefore reimposes a rhythmicity filter -- but only on the 7942 side, and it's a DATA-AVAILABILITY constraint from what Ito chose to publish, not a methodology choice made here. n: 1,846 RBH pairs -> 1,787 with expression data on both sides -> **565 with a defined peak time on both sides** (7942's own gene population is 800/2515 = 31.8% cycling by its own call, and 565/1787 = 31.6% of the ortholog subset survives the same filter -- consistent, so this isn't distorting the *rate*, just re-narrowing the set to "cycling-in-7942" specifically).

Then ran the actual requested analysis on those 565 pairs: circular $\Delta\phi$ (wrapped to $\pm 12h$), 4h-binned histogram ([figures/06_phase_offset_distribution.png](#)), a from-scratch Rayleigh test (no astropy/pycircstat available -- implemented directly, Zar's higher-order p-value correction), and a Spearman check of $|\Delta\phi|$ against each pair's weaker-side amplitude PERCENTILE RANK (not raw CV -- H3 again: raw CV isn't comparable across platforms, so ranked each gene within its own platform's full distribution first, took the worse rank per pair).

Results:

- Rayleigh: $p=1.4e-7$ ($n=565$) rejects uniformity, but weakly -- mean resultant length $R\text{-bar}=0.167$ out of 1. Mean direction $-4.6h$.
- Shape: closest to **(a) unimodal with a long tail**, not (b) bimodal dawn/dusk-swap. Histogram peaks at $[-4,0)h$ (24.1%); antipodal mass ($|\Delta\phi| \geq 8h$, 31.3%) sits BELOW the uniform baseline (33.3%) -- the opposite of what a real dawn->dusk cluster-swap would produce.
- Amplitude: Spearman $\rho=+0.014$, $p=0.74$ between $|\Delta\phi|$ and weaker-side amplitude percentile -- **no relationship**. Bottom-quartile-amplitude pairs have essentially the same mean $|\Delta\phi|$ (5.61h) as everyone else (5.81h). At $n=565$ the "big offsets are just low-amplitude noise" hypothesis (motivated by the earlier ad hoc narM look) does NOT hold up.
- narM: $\Delta\phi=-12.0h$, landing exactly on the antipode (the one point on the wrap where the sign is inherently ambiguous -- noted explicitly in the histogram's axis label). Amplitude confidence solid on both sides (44th/92nd percentile) -- not a weak-signal gene, so this reads as a genuine large offset, not measurement noise. But $n=1$: can't call one gene "representative of a class" either way.
- kaiB: $\Delta\phi=-8.8h$, 98th-percentile amplitude in 7942 (as expected -- it's the clock gene) but only 27th-percentile in Anabaena (borderline low). Independently confirms the brief's own §4 stated prior ("kai: top-amplitude in 7942, flat in Anabaena") without having gone looking for it.

What I did: Extended `VersionB_AllOrthologs.ipynb` through genome fetch -> locus/protein mapping -> RBH -> full-table join -> $\Delta\phi$ histogram -> Rayleigh test -> amplitude overlay -> narM/gap1/kaiB lookup, all executed end-to-end (`data/processed/rbh_orthologs.csv`, `data/processed/all_orthologs_rbh.csv`, `figures/06_phase_offset_distribution.png`). Genome provenance logged in `data/raw/genomes/SOURCES.txt`. Also hit and fixed a separate git issue along the way: `origin/master` had diverged (someone created, then deleted, an empty stub file at the notebook's exact path directly on GitHub) -- resolved with a standard merge (net-zero on the remote side, no real conflict), and hardened `git_workflow.py`'s `save_and_push` to surface the actual git error on push failure instead of a bare traceback.

Deliverable summary (paste-ready):

The circular phase-offset distribution across 565 confirmed RBH orthologs (defined peak time on both sides) is closest to **unimodal with a long tail** (Rayleigh $p=1.4\times 10^{-7}$ rejects uniformity, but $R\text{-bar}=0.167$ is weak; mean direction $-4.6h$), not the bimodal dawn/dusk cluster-swap hypothesis — antipodal mass sits below, not above, the uniform baseline. $|\Delta\phi|$ does **not** track amplitude confidence (Spearman $\rho=+0.014$, $p=0.74$; bottom-quartile-amplitude pairs show the same mean $|\Delta\phi|$ as everyone else), so large offsets in general are not simply low-amplitude measurement noise. narM itself sits at the extreme antipodal edge ($\Delta\phi=-12.0h$) with solid amplitude confidence on both sides (44th/92nd percentile) — not obviously noise-driven, but a lone outlier at $n=1$ rather than representative of a broader class, while kaiB ($\Delta\phi=-8.8h$, 98th/27th percentile) independently reproduces the brief's own predicted kai-gene divergence.

Transfers?: The RBH-was-actually-easy result transfers directly to env-vs-reference (same "well-curated anchor genome" situation the brief already expected). The one-sided-filter-from-data-availability finding transfers to BOTH cases and is worth generalizing: whenever two data sources are joined, check whether either side silently drops non-significant/non-called records rather than reporting a null value for them -- that silently reimposes exactly the kind of selection filter this whole exercise was built to avoid, and it will not announce itself with an error.

PART 3 — RUNNING NOTES / OPEN QUESTIONS

Moved to `runningNotes.md` (2026-07-27) to keep this file focused on the hurdle register itself. Open questions, side findings (kai genes / long-form Anabaena kaiB paralog, etc.), and build notes for next time now live there.

2026-07-27 — Bulk transcriptomics averages away minority cell types

What bit: Kushige's bulk N- array is a whole-filament average, but heterocysts are only ~ 1 per $10\text{--}15$ cells ($\sim 5\text{--}10\%$ of the population). So $\sim 90\%$ of the signal in the N- array is vegetative-cell RNA and heterocyst-specific expression is diluted roughly 10-fold in the mix. This is why Kushige could NOT read heterocyst behavior off the bulk N- data and had to run a separate heterocyst-enrichment experiment (80%-heterocyst fraction) plus a hetR-null comparison to isolate genuinely heterocyst-specific genes (all1427, hesAB). A minority-cell-type signal is only visible after enrichment or via single-cell methods; in a bulk average it can be invisible even when it's strong per-cell. **What I did / implication:** Do not treat the bulk N- array as reporting heterocyst metabolism — it reports the vegetative majority. Any heterocyst-specific question must go to the enrichment dataset (a different data type; see DEFERRED thread).

Transfers?: DIRECTLY and severely to the metagenomics work. Every environmental sample is a mixed

community; a metagenome/metatranscriptome is a population average weighted by each taxon's abundance. A rhythm (or any expression pattern) in a 5–10% member is diluted ~10–20x in bulk signal and can be undetectable — while an abundant member's pattern dominates and can look like "the community's" behavior when it's one taxon's. Quantify each taxon's fractional abundance BEFORE interpreting any bulk pattern, and treat "signal absent in bulk" as "absent OR diluted below detection," never as "absent." This is the mixed-population version of the CV/normalization problem: low apparent amplitude can be a dilution artifact, not biology. Enrichment, binning to per-MAG resolution, or single-cell/single-taxon methods are the analogs of Kushige's heterocyst fraction.